

Zhuolu Gao

Frankfurt School of Finance & Management
Adickesallee 32–34
60322 Frankfurt am Main, Germany

z.gao@fs.de
+45-91958223
www.zhuolugao.com

ACADEMIC POSITION

Postdoctoral Researcher	2025–2027
Frankfurt School of Finance & Management	

RESEARCH INTERESTS

Empirical Asset Pricing, Financial Intermediation, Credit Risk

EDUCATION

Ph.D. in Finance	2020–2025
Copenhagen Business School	
Visiting Ph.D. Student, University of Zurich (Spring 2023)	
MSc in Advanced Economics and Finance	2018–2020
Copenhagen Business School	
BSc in Finance	2011–2015
Southeast University	

JOB MARKET PAPER

The On-the-Run Discount, with Sven Klingler

The on-the-run premium is one of the most widely studied phenomena in fixed-income markets. Contrary to this phenomenon, we document that the most recently issued (on-the-run) Treasury bills trade at a *discount* relative to older bills of comparable maturities. We link this *on-the-run discount* to auction reopenings and short sellers' preference for seasoned bills. The preference arises because short sellers can cover their positions with newly auctioned bills. In line with this mechanism, short-selling demand increases as a bill ages and explains up to 67% of the variation in the on-the-run discount. Price-sensitive investors respond to the discount, but their investments are insufficient to correct it.

Award: NFA Best Paper Award in Asset Pricing

Presentations: Aarhus Finance Forum 2026, Armenian Economic Association Conference 2026*, Bundesbank*, Copenhagen Business School, FMA Annual Meeting 2026*†, FMA European Conference 2026, Frankfurt School of Finance & Management, NFA Annual Meeting 2026

WORKING PAPERS

Why Do Firms Pay More for Bank Loans? The Role of Renegotiation, solo-authored [SSRN]

Firms borrowing from both banks and the corporate bond market pay a substantial premium on bank loans, raising questions about firms' bargaining power and banks' competition. In this paper, I show that a large portion of this premium compensates banks for facilitating out-of-court restructurings. I estimate the loan premium and use a 2014 U.S. court ruling, which impeded out-of-court restructurings, as a

natural experiment. Following the ruling, affected firms experienced an 80–90 bps reduction in the loan premium, due to reduced restructuring opportunities and a diminished potential to avoid bankruptcy costs. These findings suggest that the renegotiation flexibility provided by banks is a key driver of the loan premium, highlighting the unique value that bank lending offers beyond the capital market.

Award: The Brattle Group Ph.D. Candidate Awards For Outstanding Research

Presentations: AFA poster 2025, BIGFI Research Retreat, Erasmus School of Economics, FIRS PhD Session 2024, Fudan University, Nordic Finance Network PhD Workshop 2024, NFN Young Scholars Workshop, SFI PhD Workshop, Shanghai Advanced Institute of Finance, Tri-City Day-Ahead Workshop 2024, WFA 2025

Bank Equity Risk, with Jens Dick-Nielsen and David Lando [SSRN]

Financial regulation has led banks to increase their equity ratios. Yet several studies find that this has not led to a decrease in bank equity risk. We show theoretically, that holding less capital in excess of the minimum capital requirement can outweigh the risk-reducing effect of increased total capitalization on equity. Using times series data and a natural deregulation experiment we find that excess capitalization is a significant determinant of equity risk, and can explain why bank equity risk has not become lower after the Global Financial Crisis. Lower leverage has, however, reduced the cost of bank debt.

Presentations: Aalto*, Aarhus University*, Bank of England*, Bloomberg*, CICC 2023, Copenhagen Business School, Danish Finance Institute*, Deloitte*, EFA Annual Meeting 2023*, European Banking Authority*, European Central Bank*, Luiss Finance Workshop 2026*, New York Fed*, Norges Bank, Swedish House of Finance, University of Zurich

Forgiven but Not Forgotten: Emerging Market Credit Spreads Following Debt Relief, with Mikkel Vittrup Hauerberg, David Lando, and Aleksander Koldborg Tetzlaff [SSRN]

We examine yield spreads of government debt issues from countries that have received large-scale debt relief through the Heavily Indebted Poor Countries (HIPC) program and the Multilateral Debt Relief Initiative (MDRI). Using data from more than 3,000 bond issues and after controlling for macroeconomic, political, and geographical factors, we find that HIPC governments pay an average yield spread premium of close to 1.5% on USD-denominated and non-USD-denominated bonds compared to similar countries which have not received relief. Markets seem to rationally anticipate a deterioration in the credit quality of relieved countries.

R&R at: Journal of Money, Credit and Banking

Presentations: BI Oslo (online)*, Copenhagen Business School, EFA Annual Meeting 2020 (online)*, ESADE Business School (online)*

(* presented by co-authors, † scheduled)

TEACHING

Financial Intermediation (Masters): 2021, 2022, 2024

Quantitative Methods (Masters): 2021, 2022, 2023, 2024

Master Thesis (Main Supervisor): 2024, 2025

OTHER EXPERIENCES

China Construction Bank

LANGUAGES

Mandarin (Native), English (Fluent)

PROGRAMMING

R, Python, MATLAB, Bloomberg, SAS

HONORS AND SCHOLARSHIPS

NFA Best Paper Award in Asset Pricing	2026
The Brattle Group Ph.D. Candidate Awards For Outstanding Research	2025
AFA PhD Travel Grant	2024
EFA PhD Travel Grant	2024
FIRS PhD Travel Grant	2024

REFERENCES

David Lando

Professor of Finance
Copenhagen Business School
dl.fi@cbs.dk

Jens Dick-Nielsen

Professor of Finance
Copenhagen Business School
jdn.fi@cbs.dk

Sven Klingler

Associate Professor of Finance
Frankfurt School of Finance & Management
s.klingler@fs.de